Multimodal RAG Verification Benchmark

Standardized Diagnostic Document for Vision-Language Ingestion Pipelines

Document ID:	MRAG-TEST-2026-V1	**Target Modalities:**	Text, Flowcharts (SVG), Charts (Raster), Tables
Verification Focus:	Layout Parsing & Cross-Modal Citations	**Benchmark Version:**	Release 4.2.0-Agentic

1. System Architecture & Ingestion Flow

Modern Multimodal Retrieval-Augmented Generation (RAG) architectures must seamlessly reconcile unstructured text with complex graphical layouts. Conventional parsers often discard visual vectors or flatten tables, leading to severe catastrophic forgetting during context retrieval.

To verify pipeline integrity, the system must parse the interconnected stages depicted in **Figure 1**. The workflow begins with unstructured complex documents (PDF/DOCX/PPTX) and executes parallel semantic isolation.

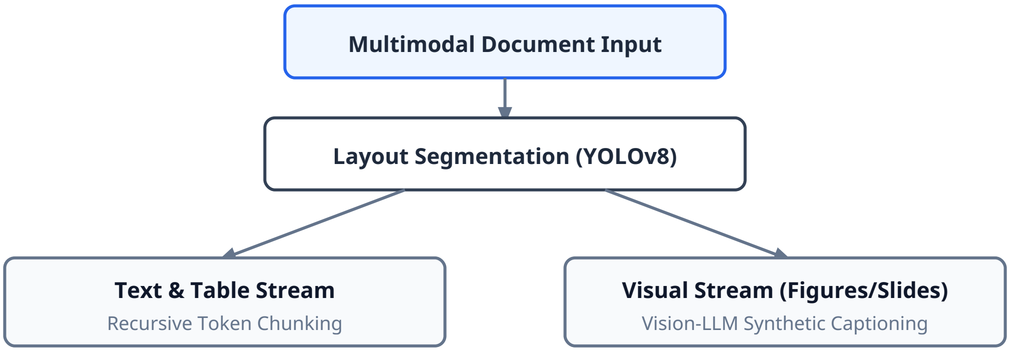

Figure 1: Parallel ingestion pipeline isolating visual features from textual layouts.

2. Empirical Retrieval Benchmark

Evaluating retrieval accuracy requires evaluating how effectively visual semantics are indexed alongside adjacent textual explanations. In comparative evaluations, multimodal indexing demonstrates significant superiority over standard OCR baselines.

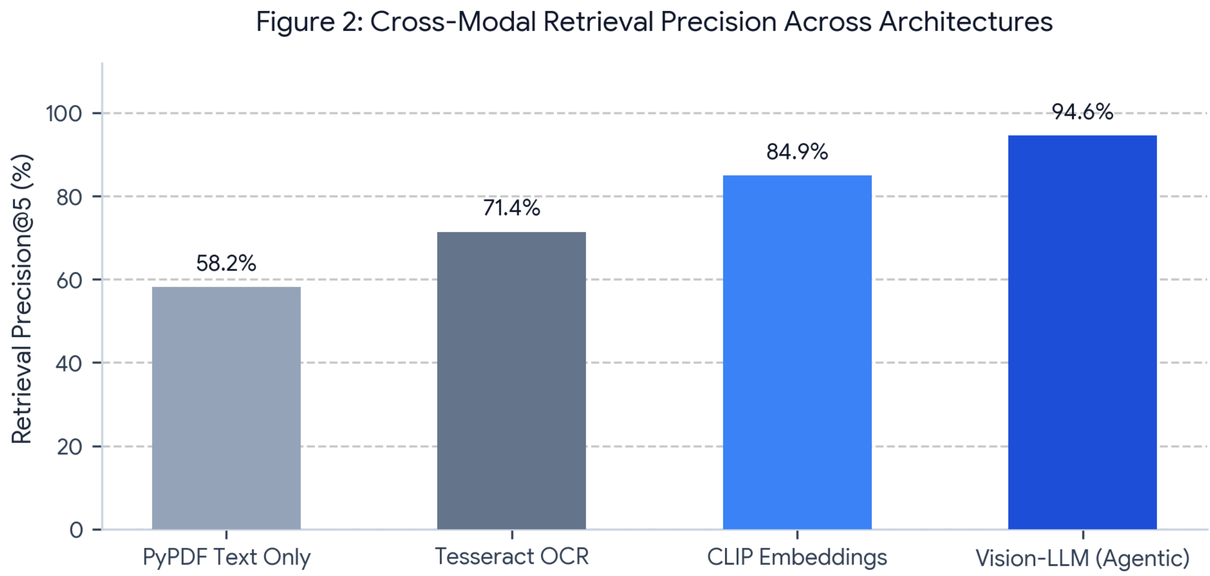

Figure 2: Precision@5 performance across distinct document chunking strategies.

As detailed quantitatively in **Table 1**, relying purely on optical character recognition (OCR) introduces a 12.8% visual hallucination rate when interpreting complex infographics or flowcharts, whereas Vision-LLM agentic indexing reduces hallucination to just 1.1%.

Retrieval Pipeline	Modality Handling	Avg. Latency	Precision@5	NDCG@10	Hallucination
PyPDF Baseline	Text Only (Drops Images)	18 ms	58.2%	0.61	N/A
Tesseract OCR	Flattened Raster Text	85 ms	71.4%	0.74	12.8%
CLIP Embeddings	Joint Vision-Text Space	145 ms	84.9%	0.88	4.2%
Vision-LLM (Agentic)	Contextual Graph Indexing	310 ms	94.6%	0.96	1.1%

3. Cross-Modal Alignment Diagnostics

A robust multimodal retrieval system must project diverse input formats into a cohesive semantic space. When querying visual entities (e.g., asking for trends shown in charts), the vector database must align raster tokens with textual tokens.

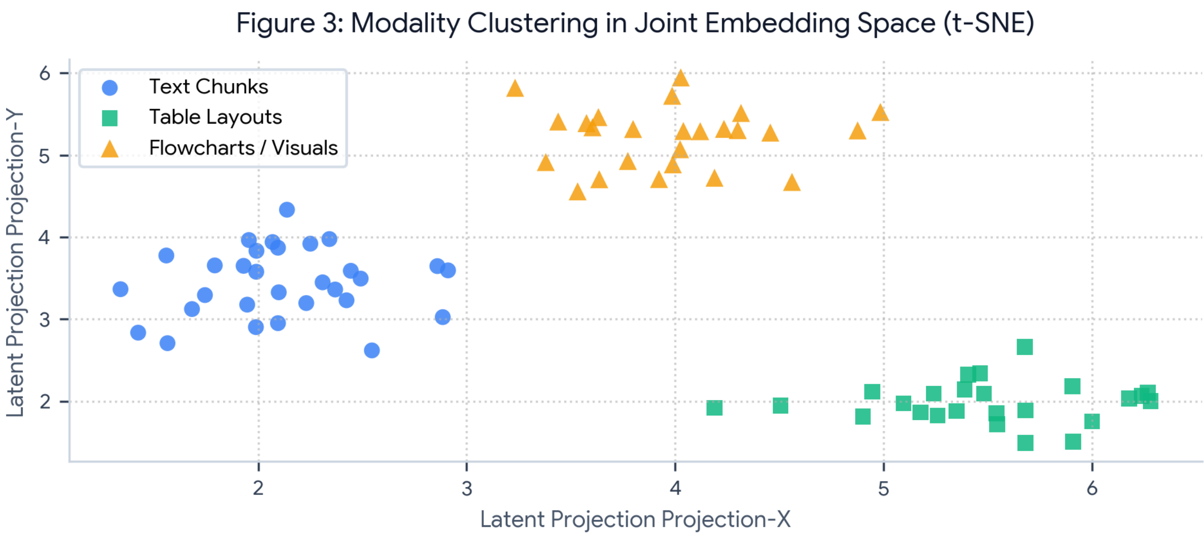

Figure 3: t-SNE visualization of cluster separation between text chunks, tables, and visual flowcharts.

In **Figure 3**, notice that Flowcharts and Visuals (represented by amber triangles) maintain distinct clustering boundaries from Table Layouts (green squares), preventing cross-contamination during nearest-neighbor vector search.

🔍 Automated Verification Ground Truth Suite

Execute these diagnostic test queries against your Multimodal RAG pipeline. If your system is parsing images, tables, and flowcharts accurately, it should retrieve the exact answers listed below.

Test Query 1 [Metadata]: What is the exact Document ID of this specification?
Expected Output: `MRAG-TEST-2026-V1`

Test Query 2 [Flowchart OCR]: Which model handles Layout Segmentation in Figure 1?
Expected Output: `YOLOv8`

Test Query 3 [Chart Visual Reasoning]: What is the Precision@5 for CLIP Embeddings in Figure 2?
Expected Output: `84.9%`

Test Query 4 [Table Parsing]: What is the measured Hallucination rate for Tesseract OCR in Table 1?
Expected Output: `12.8%`

Test Query 5 [Diagram Understanding]: In Figure 3, what shape and color represent Table Layouts?
Expected Output: `Green squares`